DVN36104 SPR 2026

Data Visualisation and Narratives
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UTS

36104 · DATA VISUALISATION AND NARRATIVES

Color and Perception

How many colours are there?

an English speaker’s basic colour words	11
the classic Crayola box	64
the web’s named colours	148
codes on a modern screen	16,777,216
differences a human can distinguish	a few million
the spectrum	a continuum

The web’s named colours

The 148 CSS colour keywords, inherited from the graphical display protocol X11 (recognised by matplotlib).

aliceblue antiquewhite aqua aquamarine azure beige bisque black blanchedalmond blue blueviolet brown
burlywood cadetblue chartreuse chocolate coral cornflowerblue cornsilk crimson cyan darkblue darkcyan
darkgoldenrod darkgray darkgreen darkgrey darkkhaki darkmagenta darkolivegreen darkorange darkorchid
darkred darksalmon darkseagreen darkslateblue darkslategray darkslategrey darkturquoise darkviolet
deeppink deepskyblue dimgray dimgrey dodgerblue firebrick floralwhite forestgreen fuchsia gainsboro
ghostwhite gold goldenrod gray green greenyellow grey honeydew hotpink indianred indigo ivory khaki
lavender lavenderblush lawngreen lemonchiffon lightblue lightcoral lightcyan lightgoldenrodyellow
lightgray lightgreen lightgrey lightpink lightsalmon lightseagreen lightskyblue lightslategray
lightslategrey lightsteelblue lightyellow lime limegreen linen magenta maroon mediumaquamarine mediumblue
mediumorchid mediumpurple mediumseagreen mediumslateblue mediumspringgreen mediumturquoise
mediumvioletred midnightblue mintcream mistyrose moccasin navajowhite navy oldlace olive olivedrab orange
orangered orchid palegoldenrod palegreen paleturquoise palevioletred papayawhip peachpuff peru pink plum
powderblue purple rebeccapurple red rosybrown royalblue saddlebrown salmon sandybrown seagreen seashell
sienna silver skyblue slateblue slategray slategrey snow springgreen steelblue tan teal thistle tomato
turquoise violet wheat whitesmoke yellow yellowgreen

English has eleven basic colour terms

black	white	red	green
yellow	blue	brown	purple
pink	orange	grey	

A basic term is a single word, not derived from an object (unlike “gold” or “turquoise”), and applies to a broad class of things.

Priority of Colour Words

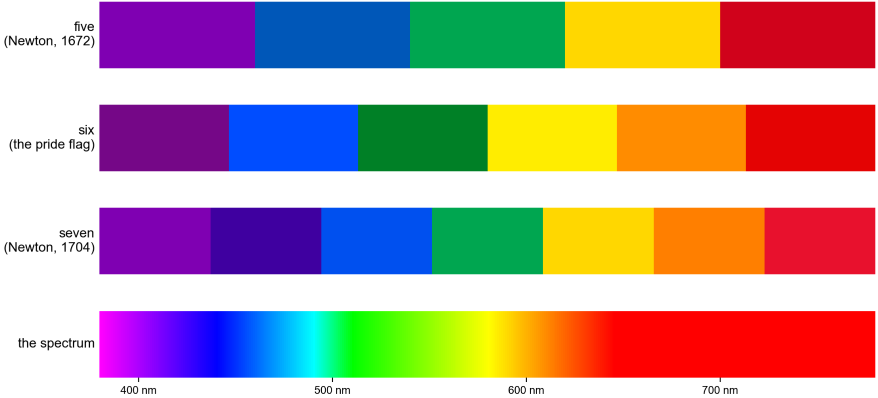
Stage	Terms present
I	black white
II	+ red
III–IV	+ green and yellow, in either order
V	+ blue
VI	+ brown
VII	+ purple pink orange grey

Languages differ in how many basic terms they have; acquisition follows this order with few exceptions. Red is the first hue named everywhere.

The classic 64-colour Crayola box

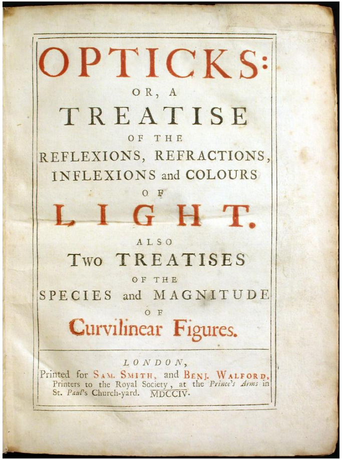
Apricot
Asparagus
Bittersweet
Black
Blue
Blue Green
Blue Violet
Bluetiful
Brick Red
Brown
Burnt Orange
Burnt Sienna
Cadet Blue
Carnation Pink
Cerulean
Chestnut
Cornflower
Forest Green
Gold
Goldenrod
Granny Smith Apple
Gray
Green
Green Yellow
Indigo
Lavender
Macaroni and Cheese
Magenta
Mahogany
Mauvelous
Melon
Olive Green
Orange
Orchid
Pacific Blue
Peach
Periwinkle
Plum
Purple Mountains' Majesty
Raw Sienna
Red
Red Orange
Red Violet
Robin's Egg Blue
Salmon
Scarlet
Sea Green
Sepia
Silver
Sky Blue
Spring Green
Tan
Tickle Me Pink
Timberwolf
Tumbleweed
Turquoise Blue
Violet (Purple)
Violet Red
White
Wild Strawberry
Wisteria
Yellow
Yellow Green
Yellow Orange

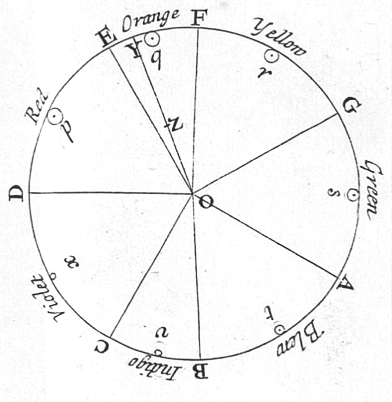
Colours of the rainbow



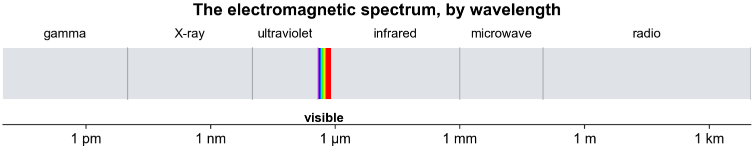
The band is continuous; the number of named colours is a choice. Newton first cut it into five, then seven, to match the notes of the musical scale.

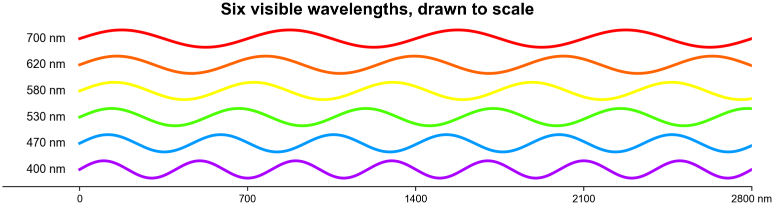
Newton, Opticks (1704)





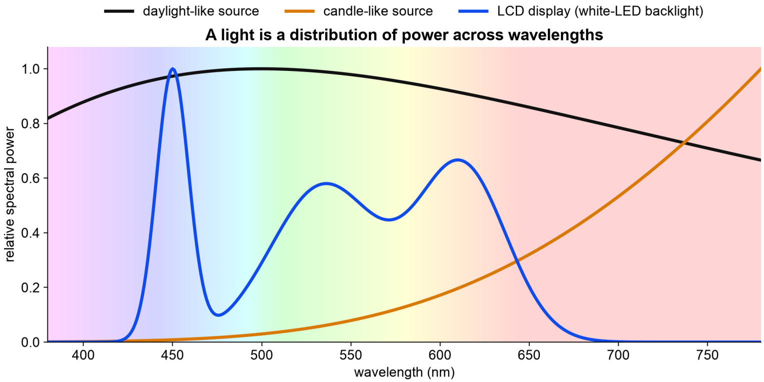
Visible light is a narrow band



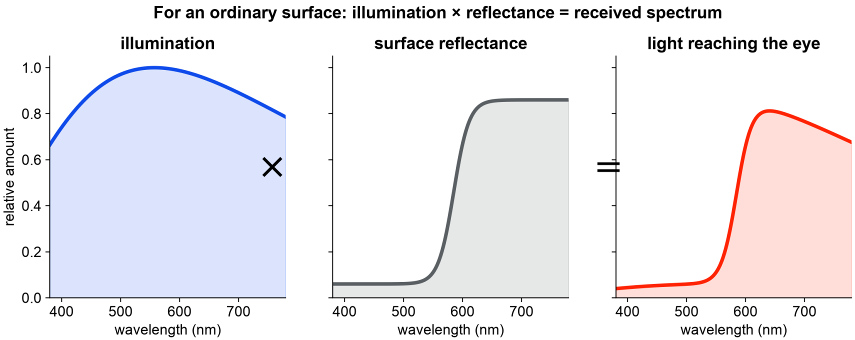


Light is electromagnetic radiation. The visible band runs from about 380 to 780 nanometres, between ultraviolet and infrared.

Spectral recipes of common lights



Light and surfaces



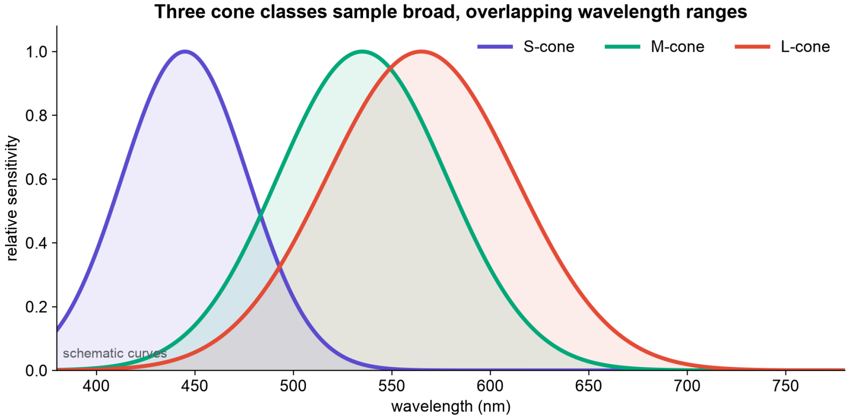
A surface reflects a different fraction of each wavelength. Its **reflectance** $R(\lambda)$ combines with the illumination $P(\lambda)$, so the light reaching the eye is $P(\lambda)R(\lambda)$. Change the light and the same surface sends a different spectrum.

Rods and cones

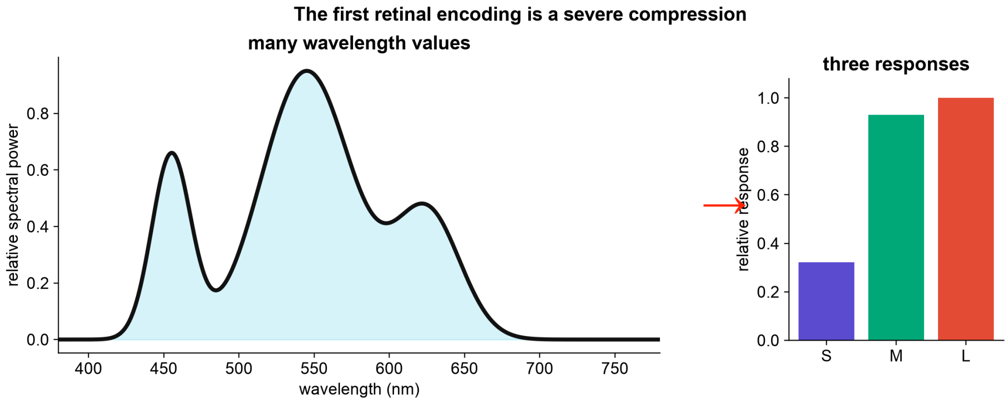
	Rods	Cones
count	about 120 million	about 6 million
work in	dim light	daylight
types	one	three
colour signal	none	three compared responses
concentration	periphery	packed in the fovea

At chart-reading light levels the rods are saturated.

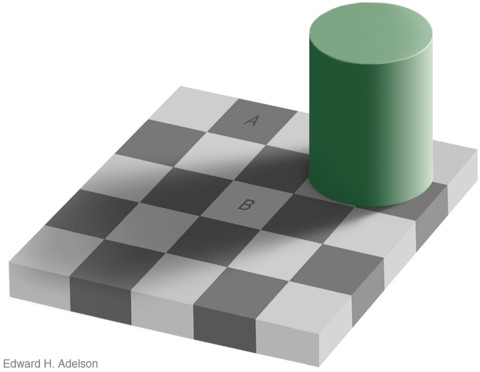
Three cone classes sample overlapping ranges



A spectrum is compressed to three responses



The same code can look different



Squares A and B are both `RGB (120, 120, 120)` or `#787878`. B looks lighter because the image makes it appear to be a light square in shadow.

After the cones, signals are compared

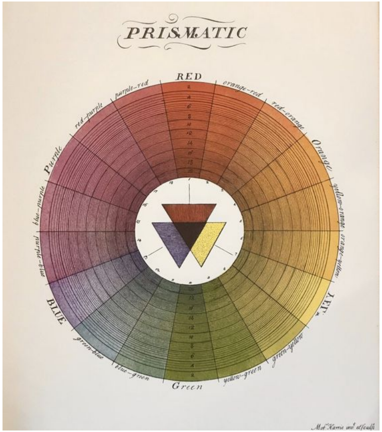
luminance-like	$L + M$
red–green-like	$L - M$
blue–yellow-like	$S - (L + M)$

Adaptation, local contrast, spatial scale, and viewing sequence all enter before conscious appearance. These are schematic combinations, far from a complete neural model. They show that the brain does not receive an RGB image from the retina.

Eight questions about colour, 1704 to 1931

Work	Object	Question
Newton, 1704	rays and prisms	How does white light separate and recombine?
Harris, 1766	pigment mixtures	How can pigment mixtures be ordered on one wheel?
Dalton, 1794	his own vision	How can colour blindness be described and explained?
Goethe, 1810	appearance	How do boundaries, shadows, afterimages, and context affect experience?
Maxwell, 1850s	matches	How can three primaries reproduce a match?
Hering, 1878	structure of appearance	Which hues look unmixed, and which blends never appear?
Munsell, 1905	surface samples	How can hue, value, and chroma be ordered for practice?
CIE, 1931	standard observer	How can colour matches be reported reproducibly?

Harris, 1766

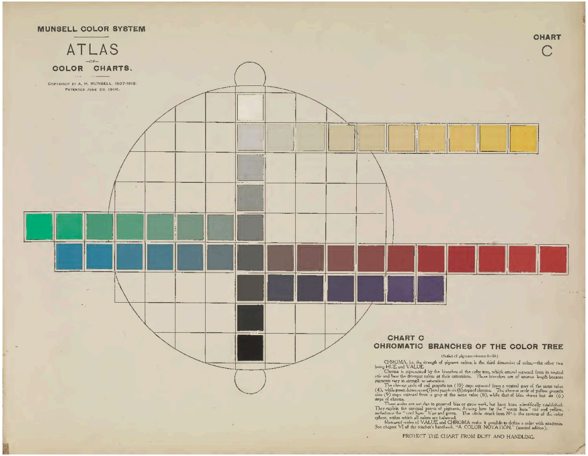


Moses Harris, an engraver and entomologist, published the first known colour circle presented in full hue in *The Natural System of Colours*. The plate was coloured by hand. Red, yellow and blue sit at the centre as the “grand or principal” colours, with mixtures grading around the circle.

Four Unique Hues (Hering, 1878)

1. Some hues look like mixtures: orange looks like red plus yellow.
2. Four do not: red, green, yellow and blue look unmixed.
3. Some blends never appear: nothing ever looks reddish-green or yellowish-blue.
4. So appearance runs on two hue axes, red–green and yellow–blue, with an unmixed pole at each end.

Munsell, 1905

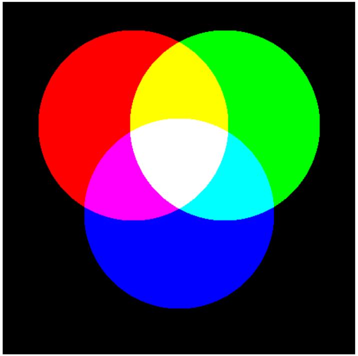


Munsell ordered painted samples along three dimensions: hue around a circle, value (lightness) up the trunk, chroma outward along the branches. Steps were chosen to look equal to observers.

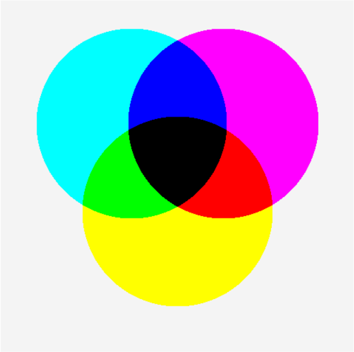
The branches have unequal lengths because pigments reach different maximum chroma at each hue and value: on this chart red extends ten steps, blue only six.

Additive and subtractive mixing

Add light: channel powers add



Stack ideal filters: transmissions multiply



Additive mixing combines emitted spectral power. Ideal subtractive mixing multiplies transmissions; real pigments also scatter and interact.

Commission Internationale de l’Éclairage

The **Commission Internationale de l’Éclairage** (CIE), or International Commission on Illumination, is the international organisation that sets standards for measuring light and colour.

Its 1931 standard observer was based on two independent colour-matching studies. W. D. Wright measured ten observers in 1928–29; John Guild measured seven in 1931. Their observers adjusted three primary lights until the mixture matched a test light. Wright used spectral primaries near **650**, **530** and **460** nm; Guild used filtered red, green and blue lamps.

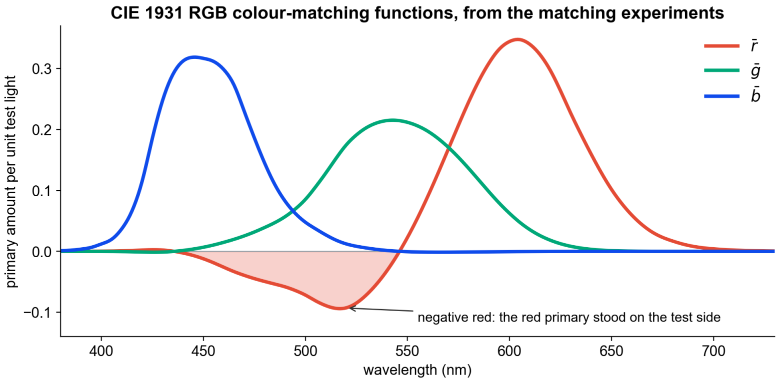
The CIE converted both datasets to a common RGB reference: **700**, **546.1** and **435.8** nm. The green and violet-blue wavelengths were reproducible emission lines from a mercury-vapour lamp. At 700 nm, small wavelength errors make little difference to the perceived red.

Why a pure cyan cannot be matched

	S	M	L
target: pure cyan, **490** nm	6	10	**4**
green lamp, **546** nm, per unit	0	10	**8**
blue lamp, **436** nm, per unit	10	1	0.5
red lamp, **700** nm, per unit	0	0	2

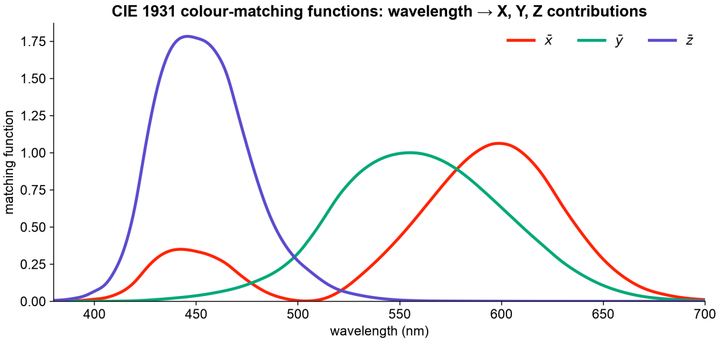
Illustrative numbers. Reaching M = 10 takes one unit of green, and that unit already delivers L = 8. The cyan needs L = 4, with the red knob already at zero. The green lamp is also a red-cone lamp. One move remains: add red to the *test* side until its L rises to meet the mix. The result is recorded as a negative red coefficient; it is not “negative light.”

Negative red



Each curve records how much of one primary matched a unit of single-wavelength test light. In the negative region the red primary stood on the test side of the field.

CIE XYZ is a mathematical coordinate system



XYZ is a linear transform of the curves on the previous slide, chosen so that no value is negative. X, Y, and Z are not physical lamps and not the three cone signals; Y was chosen to carry standard photopic luminance. The second bump of $\bar{x}$ in the violet is real: violet also excites the red-sensitive mechanism, which is why violet looks reddish.

From a spectrum to CIE values

$$X = \int P(\lambda)\bar{x}(\lambda)\,d\lambda \qquad x = \frac{X}{X+Y+Z}$$

$$Y = \int P(\lambda)\bar{y}(\lambda)\,d\lambda \qquad y = \frac{Y}{X+Y+Z}$$

$$Z = \int P(\lambda)\bar{z}(\lambda)\,d\lambda \qquad z = \frac{Z}{X+Y+Z} = 1 - x - y$$

- Spectra with the same XYZ values are a standard-observer match.
- Scaling a spectrum scales X, Y and Z together, leaving x, y and z unchanged: xy keeps the chromaticity and drops the overall scale.

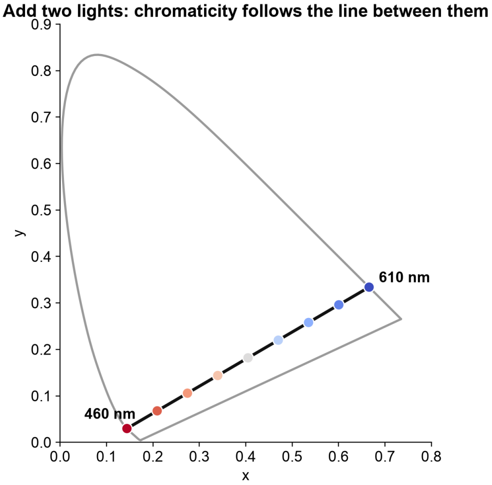
Build the spectral locus

For each row in the official CIE table:

1. take the wavelength's $\bar{x}$, $\bar{y}$, and $\bar{z}$ values;
2. divide by their sum;
3. plot (x, y);
4. repeat for the next wavelength.

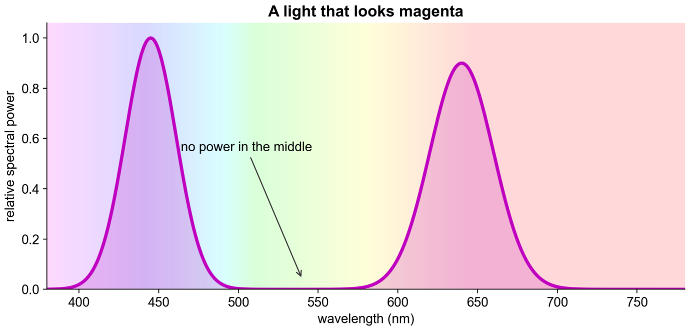
The curved path is the **spectral locus**: the route taken by monochromatic lights through the xy diagram.

Adding two lights traces a straight line



- spectra add, so XYZ values add;
- after normalisation, a two-light mixture lies on the line between the two chromaticities;
- a spectrum with many wavelengths is a non-negative mixture of many points, so every real light lands inside the locus.

Nonspectral colours



The straight edge joining the two ends of the horseshoe is the **line of purples**; no single wavelength sits on it. A light that looks magenta holds power near both ends of the spectrum, because its cone signature (S and L strong, M weak) is one no single wavelength can produce.

RGB and CMYK

	RGB	CMYK
where	screens	ink on paper
mixing	additive: channels add light	subtractive: inks absorb light
primaries	red, green, blue	cyan, magenta, yellow, black
all zero	black (screen off)	white (bare paper)
all full	white	muddy near-black
you meet it as	`#RRGGBB` in CSS and charts	print exports and PDF proofs

The K in CMYK is a separate black ink, the printers' "key" plate. Full cyan, magenta and yellow together make a muddy brown, so text and shadows use black ink instead. Screens and printers also cover different gamuts: some screen colours have no printable match.

Metamerism

Two lights with different spectra that produce the same three cone responses are **metamers**: they look identical. A screen emits a new spectrum from three primaries, built to be a metamer of the colour it is asked to show.

What this makes possible

- colour photography;
- television and computer displays;
- compact three-channel specifications.

Where a match can fail

- a different observer;
- a different illuminant;
- fluorescence;
- changed adaptation or field size.

Bits per pixel

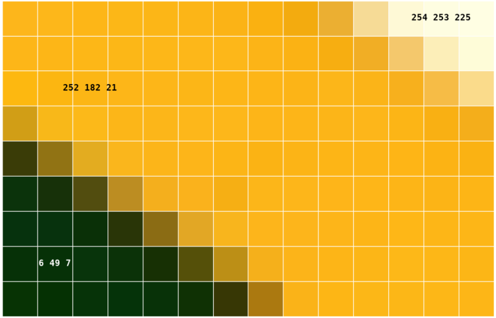
Bits per pixel	Colours	Era
1	2	terminals, early Macintosh
2	4	CGA graphics, 1981
4	16	EGA and VGA high-resolution modes
8, indexed	256	VGA 320×200 mode, 1987
15 or 16	32,768 or 65,536	high colour, early 1990s
24	16,777,216	true colour, since the mid-1990s

24 bits is the `#RRGGBB` code: two hexadecimal digits per channel. Chart libraries, CSS and image formats all speak it.

A picture is a grid of numbers

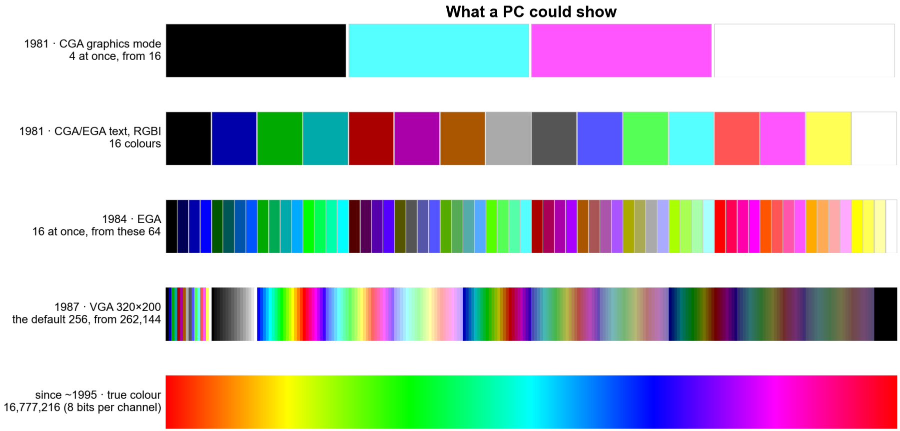
the picture

the marked region: 14 × 9 pixels



Zoom far enough into any picture and it becomes a grid of squares: **pixels**. A **bitmap** is that grid stored as numbers, three per pixel: how much red, green and blue light to emit, each from 0 to 255.

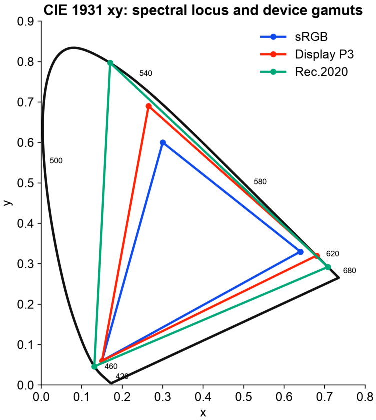
What a PC could show



The default 16-colour VGA palette

Black (0, 0, 0) #000000	**Blue** (0, 0, 170) #0000AA	**Green** (0, 170, 0) #00AA00	**Cyan** (0, 170, 170) #00AAAA
Red (170, 0, 0) #AA0000	**Magenta** (170, 0, 170) #AA00AA	**Brown** (170, 85, 0) #AA5500	**Light grey** (170, 170, 170) #AAAAAA
Dark grey (85, 85, 85) #555555	**Light blue** (85, 85, 255) #5555FF	**Light green** (85, 255, 85) #55FF55	**Light cyan** (85, 255, 255) #55FFFF
Light red (255, 85, 85) #FF5555	**Light magenta** (255, 85, 255) #FF55FF	**Yellow** (255, 255, 85) #FFFF55	**White** (255, 255, 255) #FFFFFF

Three physical primaries make a triangle



Colour bars

The broadcast test pattern is the gamut triangle enumerated: the three primaries alone (green, red, blue), their pairwise sums (yellow, cyan, magenta), and all three together (white).

Every other colour the screen can produce is a mixture of these, inside the triangle.

Five cautions about the xy diagram

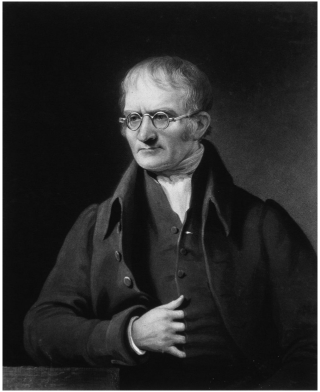
- The outline represents standard-observer chromaticity coordinates.
- The page is itself shown through your display’s primaries and calibration.
- Points outside the display gamut can be plotted as locations but not reproduced faithfully.
- Different luminances share the same xy point.
- Equal distances in xy are not equal perceived differences.

Test a palette in its context

- test colours on the actual background;
- test them at the actual mark and text sizes;
- inspect common colour-vision deficiencies;
- make lightness monotonic in sequential scales;
- reserve diverging palettes for a meaningful midpoint;
- add labels, position, shape, or line style when the distinction matters.

What matters is how the marks look next to each other on the actual page.

Dalton, 1794



Dalton gave the first scientific self-report of colour blindness: to him, “that part of the image which others call red appears to me little more than a shade”. He blamed a blue tint in his eye’s fluid.

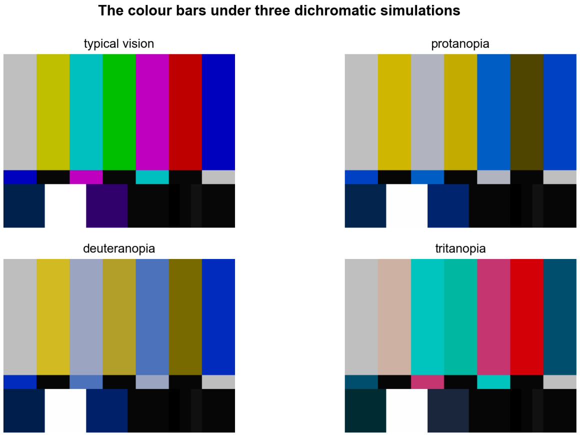
He was wrong about the cause. DNA from his preserved eye (1995) showed he lacked the M-cone pigment: one of the three integrators from Part II was missing, so his matches needed only two primaries.

Colour vision deficiencies

Type	Affected pigment	Rate
deutan	M shifted or missing	about 6% of men
protan	L shifted or missing	about 2% of men
tritan	S affected	about 1 in 10,000, both sexes
monochromacy	two or three pigments	very rare

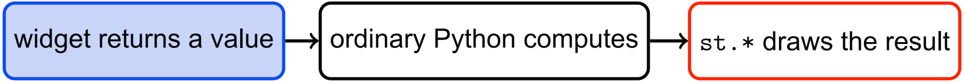
The red–green forms are X-linked, so they concentrate in men: about 8% of men and 0.5% of women. In a class of two hundred students, expect eight or nine.

The colour bars, simulated



Streamlit

How a Streamlit app runs



Whenever a widget changes, Streamlit reruns the script from top to bottom with the new value.

A slider replaces a constant

Notebook or script

```
wavelength = 550
print(wavelength)
```

Streamlit

```
import streamlit as st
wavelength = st.slider(
    "Wavelength (nm)", 380, 700, 550
)
st.write(wavelength)
```

```
python -m streamlit run hello_colour.py
```

The lab builds three linked views

1. **Spectrum**: change peak wavelength and spectral width.
2. **Eye**: integrate the spectrum against three schematic cone curves.
3. **Chromaticity**: use the official CIE table to locate the light in xy.

Final extension: add an intensity slider. Cone-response magnitudes should change; chromaticity should remain almost fixed.

The app's simplifications

- Cone curves in the app are schematic.
- CIE matching functions are standard-observer data, not cone sensitivities.
- A wavelength rendered on screen is an RGB approximation, not monochromatic light.
- xy omits luminance and viewing context.
- The app is an explanatory model, not a calibrated colour-management system.

Sources and data

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